Guideline on Tool Confidence Level Determination According to ISO 26262

Auxiliary Information

DaVinci Tools

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Document Information

Approval

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Review

Date Name (Role)
2017-08-21, Jonas Wolf (Safety Manager)

History

Version	Remarks
2.1	Initial version, combined document for DaVinci DEV and DaVinci CFG PRO (Matthias Wernicke)
2.2	Review (Jonas Wolf)
2.2.1	Minor correction in product names (Matthias Wernicke)
3.0	MSSV and RTE Analyzer added (Matthias Wernicke)
3.1	Win8.1 and Win10 added
4.0	Additional use case “issue relevance analysis”
4.0.1	Formal change
4.1.0	New DaVinci product names added (Matthias Wernicke)

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1 Introduction

1.1 Purpose of This Document

This document provides users of the **DaVinci Tools** with recommendations how to determine their Tool Confidence Level (TCL) according to the methodology described in ISO 26262 – Part 8. Use cases relevant for the development of safety-related systems are analysed. For each use case, a classification of the Tool Confidence Level (TCL) is suggested. If qualification methods are needed a recommendation is given for qualification methods. Wherever relevant, requirements and assumptions on the context in which the tool is used that have an impact on the TCL are stated.

1.2 Product Identification

The recommendations within this document apply for the following products and options:

DaVinci Configurator Pro 5.x and DaVinci Configurator Classic 5.x including all service packs (in this document referred to as “DaVinci Configurator”)

DaVinci Developer 3.x, DaVinci Developer 4.x and DaVinci Developer Classic 4.x including all service packs (in this document referred to as “DaVinci Developer”)

1.3 Product Goals and Usage

DaVinci Configurator is a software tool for configuring the BSW and the RTE of an ECU. The tool is used by ECU SW integrators to configure the BSW and the RTE of an AUTOSAR ECU. For this purpose, the tool provides a graphical and text-based UI to create AUTOSAR-compliant XML files with the BSW and RTE configuration.

DaVinci Developer is a software tool for designing SWCs and for configuring the RTE of an ECU. The tool is used by ECU SW developers to design the structure and interface of the ECU’s application software. For this purpose, the tool provides a graphical and text-based UI to create AUTOSAR-compliant XML files including the SWC description. Furthermore, the tool provides a validation function to check the consistency and AUTOSAR compliance of the SWC descriptions.

The tools are used in combination with MICROSAR code generators, and external compiler and linker programs. In this combination, the tools provide a validation function to check the consistency and AUTOSAR compliance of the BSW and RTE configuration, and to generate embedded code of the BSW modules and the RTE.

For details concerning features, see the online help of the tools.

2 Use Cases

2.1 Configure and Generate MICROSAR SafeBSW Modules and/or SafeRTE

Intended Purpose:	Configure BSW modules and/or RTE of an AUTOSAR ECU and generate embedded code
Environmental, Functional and Process Constraints:	Context for the use case > Operating system: Windows 7 SP1 (64bit) or Windows 8.1 (64bit) or Windows 10 (64bit) > DaVinci Configurator software tool as mentioned in chapter 1.2 > Optionally: DaVinci Developer software tool as mentioned in chapter 1.2 Usage of the DaVinci software tools in combination with MICROSAR SafeBSW and SafeRTE. Usage of the tools in combination with other BSW/RTE products is not addressed.
Description:	Interactive usage 1. Start DaVinci Configurator to setup an ECU project based on input files, or open an existing project. 2. Optionally: Create or edit the SWC descriptions using the UI of DaVinci Developer. 3. Create or edit the BSW and/or RTE configuration using the UI of DaVinci Configurator. 4. Validate the BSW and/or RTE configuration using the UI of DaVinci Configurator. 5. Run the code generation Command line usage Start DaVinci Configurator command line based on an existing project configuration and run the code generation. Note: Validation of BSW and RTE configuration is enforced end prevents code generation in case of errors.
Output:	Embedded code
Tool Impact:	TI2 (Possibility of impact on a safety requirement) Rationale: > Tool errors may lead to embedded code behavior, which deviates from the intention of the user.
Tool Error Detection:	TD1 (High degree of confidence of prevention or detection) Rationale:

	> Many potential faults, e.g. corrupt input files or partially inconsistent user input, are detected internally in the DaVinci Configurator before creating the embedded code. > Examples: XML validation, validators in DaVinci Configurator, tool internal assertions > Majority of remaining relevant faults in tool chain are discovered during an ISO 26262-compliant process of the item development including the software integration, verification of software safety requirements, item integration and safety validation steps. > Examples: Compile or link errors, wrong communication partners, delayed communication, … > All other faults in the RTE are addressed by the RTE Analyzer, because they are harder to detect during following process steps. Faults leading to memory corruption in the BSW are detected using MSSV.
Tool Confidence Level[1]:	TCL1[2]
Recommended method of qualification:	Not required

Table 2-1 Configure BSW Modules

2.2 Relevance Analysis of MICROSAR Issues

Intended Purpose:	Analyze if the issues described in a MICROSAR issue report are relevant for a particular project
Environmental, Functional and Process Constraints:	Context for the use case > Operating system: Windows 7 SP1 (64bit) or Windows 8.1 (64bit) or Windows 10 (64bit) > DaVinci Configurator software tool as mentioned in chapter 1.2
Description:	Start DaVinci Configurator command line based on an existing project configuration and run the issue report analysis
Output:	Relevance report, indicating if the issue report contains issues that are relevant for the project
Tool Impact:	TI2 (Possibility of impact on a safety requirement) Rationale: > Tool errors may lead to an incomplete relevance report. I.e., a MICROSAR issue that affects a safety requirement of the project is not recognized and reported as such.
Tool Error Detection:	TD3 (Low degree of confidence of prevention or detection) Rationale: > The incompleteness of a relevance report created by the DaVinci tools is not obvious for humans. There is no redundant tool or

[1] The TCL is determined from TI and TD. Details see Appendix.

[2] Note that this classification applies to the DaVinci products defined in section 1.2. For the tools “MSSV“ and „RTE Analyzer“ TCL2 is recommended. Further details can be found in the Safety Manual of MICROSAR Safe.

	alternative verification that is able to compensate errors in the DaVinci tools.
Tool Confidence Level[3]:	TCL3[4]
Recommended method of qualification:	Avoid usage of the DaVinci tool. Perform a manual relevance analysis of the MICROSAR Issue Report instead. Vector provides the Issue Report in human readable format, so this approach is feasible.

Table 2-2 Analysis of MICROSR Issues

[3] The TCL is determined from TI and TD. Details see Appendix.

[4] Note that this classification applies to the DaVinci products defined in section 1.2. For the tools “MSSV“ and „RTE Analyzer“ TCL2 is recommended. Further details can be found in the Safety Manual of MICROSAR Safe.

3 Summary

Use Case	TCL	Qualification Method
Configure and Generate MICROSAR SafeBSW Modules	1	Not required
Relevance analysis of MICROSAR issues	3	Avoid usage of the DaVinci tool. Perform a manual relevance analysis of the MICROSAR Issue Report instead. Vector provides the Issue Report in human readable format, so this approach is feasible.

Table 3-1 Use Cases

4 Appendix

The TCL is determined from TI and TD according to the following scheme:

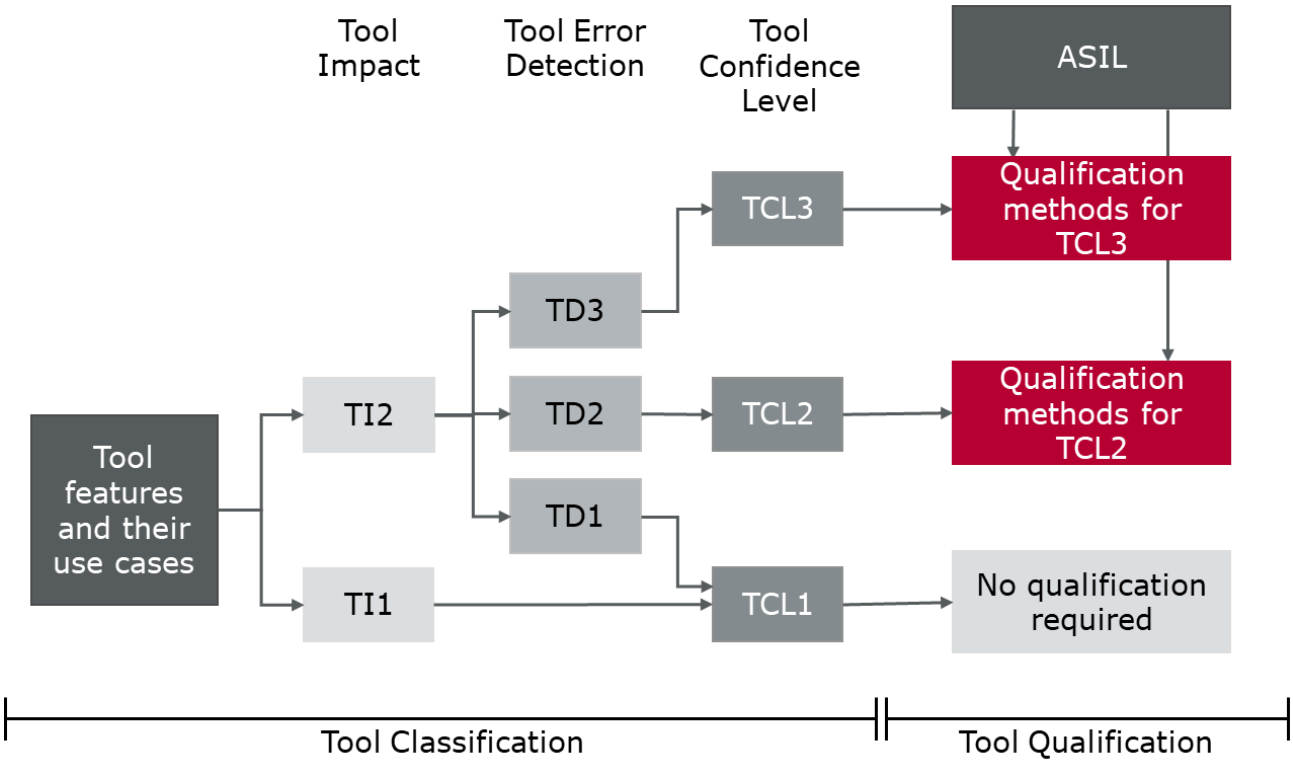


Figure 4-1 TCL Scheme

5 Contact

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